

WATTISHAM, United Kingdom

WMO: 035900

Lat: 52.1239N

Lon: 0.9575E

Elev: 87

StdP: 100.29

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.3	(c) -2.1	(d) -6.1	(e) 2.3	(f) -1.1	(g) -4.7	(h) 2.6	(i) -0.5	(j) 14.8	(k) 8.3	(l) 13.3	(m) 7.9	(n) 3.4	(o) 250	(p) 0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.2	(c) 26.5	(d) 18.6	(e) 24.5	(f) 17.7	(g) 22.8	(h) 16.8	(i) 19.5	(j) 24.7	(k) 18.5	(l) 23.0	(m) 17.6	(n) 21.5	(o) 4.7	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	12.7	21.3	16.8	12.1	20.3	16.1	11.5	19.4	55.9	24.7	52.7	23.0	49.9	21.6	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	10.4	9.3	-6.2	30.0	1.9	2.0	-7.6	31.4	-8.7	32.6	-9.8	33.7	-11.2	35.1
			-6.6	21.9	1.9	1.8	-7.9	23.2	-9.1	24.2	-10.1	25.2	-11.5	26.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	4.1	4.5	6.2	8.7	11.8	14.7	17.1	17.0	14.4	11.1	7.2	4.7
	DBStd	5.55	3.11	3.20	3.06	3.13	3.07	2.74	2.71	2.62	2.53	2.91	3.08	3.53
	HDD10.0	820	184	156	121	60	17	1	0	0	1	21	93	167
	HDD18.3	3029	442	387	375	288	204	114	57	58	121	224	336	423
	CDD10.0	877	1	1	4	22	71	141	222	218	133	56	8	2
	CDD18.3	46	0	0	0	0	1	4	20	17	3	1	0	0
	CDH23.3	305	0	0	0	1	3	34	141	111	13	2	0	0
	CDH26.7	51	0	0	0	0	0	4	24	21	1	0	0	0
(14) Wind	WSAvg	4.7	5.4	5.4	5.2	4.7	4.6	4.3	4.2	4.2	4.4	4.7	4.8	5.1
(15) Precipitation	PrecAvg	582	49	37	42	44	44	48	48	48	54	55	61	53
	PrecMax	788	133	93	95	108	103	114	101	113	119	125	156	100
	PrecMin	440	6	4	7	5	5	5	5	3	1	3	19	15
	PrecStd	91	24	19	23	24	21	29	25	29	33	34	28	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	13.7	16.0	21.3	23.6	26.9	29.2	29.0	25.0	20.3	15.5	13.4
		MCWB	10.9	10.9	10.9	13.8	16.4	18.8	19.3	19.4	17.7	16.0	13.8	11.8
	2%	DB	11.3	11.7	14.0	18.0	21.3	23.9	26.7	26.4	22.3	17.9	14.0	12.2
		MCWB	10.0	9.6	10.4	12.5	15.4	17.4	18.8	18.5	16.9	14.8	12.5	11.0
	5%	DB	10.0	10.4	12.5	15.7	19.4	21.9	24.7	24.0	20.5	16.4	12.9	11.2
		MCWB	8.8	8.6	9.5	11.1	14.2	16.4	17.7	17.6	15.8	14.1	11.5	10.0
	10%	DB	8.9	9.2	11.1	14.1	17.5	20.1	22.8	22.0	18.9	15.4	11.7	9.9
		MCWB	7.7	7.7	8.7	10.2	13.1	15.3	16.9	16.6	14.8	13.5	10.5	8.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.3	12.0	14.7	17.8	20.2	20.9	20.6	19.2	16.7	14.1	12.2
		MCDB	12.2	13.0	14.3	19.6	22.2	25.5	27.1	26.5	23.5	19.6	15.0	13.0
	2%	WB	10.1	9.9	11.0	12.9	16.1	18.4	19.6	19.2	17.7	15.5	12.8	11.3
		MCDB	11.1	11.3	13.1	17.1	20.2	22.6	25.2	24.0	21.0	17.2	13.7	12.2
	5%	WB	8.9	8.8	10.0	11.7	14.7	17.0	18.5	18.3	16.7	14.7	11.7	10.1
		MCDB	9.9	10.1	12.1	15.0	18.7	20.6	23.2	22.5	19.3	16.0	12.7	11.0
	10%	WB	7.8	7.9	9.0	10.7	13.5	15.9	17.6	17.4	15.7	13.8	10.6	9.0
		MCDB	8.8	9.1	10.9	13.5	16.9	19.4	21.6	20.9	18.0	15.1	11.6	9.9
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.6	6.9	8.3	8.6	8.8	9.2	8.9	8.0	6.6	5.4	4.7
		MCDBR	6.2	6.6	8.5	10.5	11.4	11.2	12.0	11.2	9.7	7.2	5.8	5.3
		MCWBR	5.8	5.1	5.7	6.0	6.5	6.0	5.3	4.8	5.1	4.8	5.0	4.9
	5% WB	MCDBR	6.1	6.2	8.0	9.8	10.7	10.4	10.9	10.1	8.7	6.5	5.5	5.3
		MCWBR	5.9	5.2	5.8	5.9	6.4	6.2	5.5	4.8	5.1	4.9	5.1	5.1
(40) Clear-Sky Solar Irradiance	taub		0.325	0.342	0.376	0.398	0.404	0.407	0.406	0.396	0.374	0.357	0.330	0.315
	taud		2.369	2.381	2.278	2.237	2.265	2.277	2.293	2.342	2.398	2.429	2.413	2.355
	Ebn at Noon		670	771	808	835	848	847	841	830	805	743	660	611
	Edh at Noon		61	80	106	124	127	127	123	111	94	75	58	53
(44) All-Sky Solar Radiation	RadAvg		0.75	1.42	2.54	4.02	4.84	5.22	4.98	4.18	3.13	1.81	0.92	0.59
	RadStd		0.07	0.21	0.29	0.41	0.36	0.41	0.49	0.31	0.26	0.17	0.10	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page